Rutgers University
Master's Comprehensive Examination

Part I — Theory (90 minutes) — December 1, 1990

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. Let X be a binomial random variable which is the number of successes in n independent Bernoulli trials. Let Y be the number of successes in the first $m < n$ trials. Find the conditional frequency function of $Y|X = x$ and the conditional mean.

2. An item is present in a list of n items with probability p and if it is present its position in the list is uniformly distributed. A program searches through the list sequentially. Find the expected number of items searched before the program terminates.

3. A fair coin is tossed n times and the number of heads N is counted. It is then tossed N more times. Find the expected total number of heads generated by this process.

4. Suppose X is a random variable whose density is

$$f_X(x;\theta) = e^{-(x - A)}, \qquad x > A.$$

a. Find EX.

b. Suppose X_1, X_2 are sample points from the above population. Find the m.l.e. for A and find an unbiased estimate of A.

5. Three children are selected at random (without replacement) from a group of six with ages 3, 3, 3, 4, 4, and 5. Let X be the sum of ages of selected children. Find the probability distribution of X.

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Part II — Applied (90 minutes) — December 1, 1990

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. An experiment was performed using eight tomato plants. One half of the flowers of each plant was pollinated by the experimenter using a new pollinating technique while on the other half of the plant the experimenter used a standard pollinating technique. The seed yield from each half of the plant is tabulated below.

Plant No.	Seed yield from experimental pollinating technique	Seed yield from standard pollinating technique
1	10	8
2	12	14
3	8	6
4	12	10
5	14	10
6	8	8
7	10	12
8	12	10

a. Determine if the experimental technique gives a higher yield in seed than the standard technique (let $\alpha = .05$). Be sure that both H_0 and H_1 are shown.

b. How can the experimenter make a Type I error? What are the consequences of his doing so?

c. How can the experimenter make a Type II error? What are the consequences of his doing so?

d. Give a 90% confidence interval for the difference between the two techniques.

2. a. Describe the goodness of fit test for a composite hypothesis $p_i = \pi_i(\theta_1,\ldots,\theta_m)$ $(i = 1,\ldots,k)$. What are the underlying assumptions?

b. At the 5% level, test the fit of a Poisson distribution to the following data. Here n_j is the number of observations equal to j $(j = 0,1,\ldots)$, while $n = \Sigma n_j = 300$.

$j \rightarrow$	0	1	2	3	4	5	6	7	8
$n_j \rightarrow$	6	24	42	59	62	44	41	14	8

(Continued)

3. A simple linear regression model

$$Y = \alpha + \beta X + \epsilon$$

was fit to a set of data. The following results were found.

$$b = 4$$
$$\bar{X} = 4$$
$$\bar{Y} = 20$$

Additionally part of the analysis of variance table is summarized below.

Source	df	ss	ms	F
Total	10	800		
Regression				
Residual		288		

a. Find the estimated regression line.
b. If it is assumed that ϵ is distributed as N $(0, \sigma^2)$ then estimate σ^2.
c. Test the hypothesis that $\beta = 0$. Let the probability of a Type I error = .10.
d. Find a 95% confidence interval for the mean value of Y when X = 8.
e. Find the coefficient of determination. What does this mean?
f. Find an estimation of the correlation coefficient, ρ.

4. For each of the following situations, assume that an Analysis of Variance is to be performed. List the sources of variation, the degrees of freedom and the expected mean square. Be sure to indicate which sources of variation are fixed and which are random.

a. An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. Each of 6 state experiment stations has agreed to grow the 3 varieties. Each station will homogenize a randomly selected tomato of each variety, determine the percent acid content, and submit this one measurement (for each variety) to the experimenter.

b. An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. From a single experiment station he obtains a bushel of each variety. For each variety he then selects 6 tomatoes at random from the bushel and records the percent acid content of each of the selected tomatoes.

c. An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. In a greenhouse, 4 benches were used. Placed on each bench was a single plant from each variety. From each plant 2 fruit were selected and the percent acid content of each fruit was recorded.

d. An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. He finds 6 home gardeners who are growing variety A, 6 who are growing variety B, and 6 who are growing variety C. Each gardener will supply the experimenter with 2 tomatoes and the experimenter will then record a measure of percent acid content for each tomato.

(Continued)

5. The following hypothesis was tested using a z–test with $\alpha = .05$

$$H_0: \mu = 5$$
$$H_1: \mu \neq 5$$

The z–test was based on n = 20 and $\bar{X} = 3$. From this test H_0 was rejected and H_1 was accepted.

Based on this information determine if each of the following statements is True, False, or there is insufficient information to make a determination. For each statement use:

T for True
F for False
C for cannot be determined.

In addition explain the reason for your answer.

a. If $\bar{X}$ and α remain the same and n is increased by 30 H_0 will be rejected and H_1 will be accepted.

b. If $\bar{X}$ and α remain the same and n is decreased to 10 H_0 will be accepted.

c. If $\bar{X}$ and n remain the same and α is changed to .10 then H_0 will be accepted.

d. If $\bar{X}$ and n remain the same and α is changed to .01 then H_0 will be rejected and H_1 will be accepted.

e. If n and α remain the same and $\bar{X} = 2$ then H_0 will be accepted.

f. If n and α remain the same and $\bar{X} = 4$ then H_0 will be accepted.

Repeat questions (a) through (f), call them (g) through (ℓ), respectively, except this time assume that from the z–test H_0 was accepted.

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MAKEUP MASTERS EXAMINATION, SPRING 1990

90 minutes for each part – Open book, open notes

PART I: THEORY

Answer 4 out of the following 5 questions

1) In a town with 5000 adults, a sample of 100 is asked their opinion of a proposed municipal project; 60 are found to favor it, and 40 oppose it. If, in fact, the adults of the town were equally divided on the proposal, what would be the probability of obtaining a majority of 60 or more favoring it in a sample of 100?

2) If $X_1, X_2, \ldots, X_k$ are independent Poisson distributed random variables, find the conditional distribution of X_1 given $X_1 + X_2 + \ldots + X_k$.

3) Suppose that $X_1, X_2, \ldots, X_n$ is a sample from a population with density $f(x,\theta) = \theta x^{\theta-1}$, $0 < x < 1$, $\theta > 0$.

a) Find the Fisher information $I_n(\theta)$ for θ.

b) Find an UMVUE of $1/\theta$.

4) Let $X_1, X_2, \ldots, X_n$ denote the incomes of n persons chosen at random from a certain population. Suppose that each X_i has the density

$$f(x,\theta) = c^{\theta}\,\theta\, x^{-(1+\theta)}, \quad x > c$$

where $\theta > 1$ and $c > 0$.

a) Express mean income μ in terms of θ.

b) Find the optimal test statistic for testing H_0: $\mu = \mu_0$ versus H_1: $\mu > \mu_0$.

5) Let $X_1, \ldots, X_n$ be independently and identically distributed with density

$$f(x,\theta) = \frac{1}{\sigma} \exp\{-(x-\mu)/\sigma\}, \quad x \geq \mu$$

where $\theta = (\mu, \sigma^2)$, $-\infty < \mu < \infty$, $\sigma^2 > 0$.

a) Find maximum likelihood estimates of μ and σ^2.

b) Find the maximum likelihood estimate of $P_\theta[X_1 \geq t]$ for $t > \mu$.

MAKEUP MASTERS EXAMINATION, SPRING, 1990

PART II: APPLIED

Answer 4 out of the following 5 questions

1) We seek to estimate the mean age of individuals living in a very large development. We want the estimate to be accurate to within 1 year, with 95% confidence. We know that most of the individuals are between ages 50 to 80. (You can assume that "most" means 95%, and that ages are approximately normally distributed for this development). How big a sample (simple random sample) is needed.

2) In a particular community, 40% of the adult population belong to a minority group A. A review of a random sample of 100 names on the Jury lists, reveals that only 25 out of the 100 names (or 25%) belong to the minority group A.

The questions we are interested in are the following:

a) Is there statistically significant evidence that the minority group A has a smaller chance of being selected for Jury lists, than individuals not in group A?

b) Aside from sampling variability, you can estimate from the above data that members of the non-minority group are C times as likely to be selected for Jury list as are members of the minority group A. Give (and justify) your estimate of C, to two decimal places.

3) A researcher seeks to compare six different treatments for the common cold. A completely randomized experiment is conducted, with two patients assigned to each of the treatments. A measure, y, is made of how effective is each treatment. (The higher the y, the more effective the treatment.) The Y scores are as follows:

Treatment:	1	2	3	4	5	6
	20	36	29	3	4	24
	28	30	37	1	18	10

Carry out the one-way Analysis of Variance for this data, showing complete ANOVA table with source, df, SS MS, F ratios.

4) Two panel of taste-testers were independently asked to rank seven brands of coffee from best (rank 1) to worst (rank 7). The following is the ranking of the two panels.

Brand	A	B	C	D	E	F	G
panel I	1	2	4	6	3	5	7
panel II	1	2	3	7	4	5	6

a) Compute an appropriate statistic to measure agreement between the panels.

b) Use the statistic to test whether the apparent agreement is just due to sampling variability due to small samples used.

5) A large company sought to compare the effectiveness of two different brands of typewriters on office efficiency. Each typewriter could be ordered with either a standard keyboard, or with a fancier keyboard with a separate numeric pad. The treatment combinations are thus:
T1: Brand A with standard keyboard
T2: Brand A with augmented keyboard
T3: Brand B with standard keyboard
T4: Brand B with augmented keyboard.

The company has four separate locations, and at each location has four separate divisions: accounting, scientific, sales, manufacturing. Within a given division at a particular location, only one type of typewriter can be used. Because of this, the company chose a Latin Square Design as follows:

Location	Division			
	acctng	science	sales	mftg
N.Y.	T1	T2	T3	T4
Philadel.	T4	T1	T2	T3
Washington	T3	T4	T1	T2
Detroit	T2	T3	T4	T1

The Mean and Total scores for the four treatments are as follows:

Treatment	T1	T2	T3	T4
Mean	5	6	8	12
Total	20	24	32	48

Fill in the following Anova Table

Source	df	SS	MS
Treatment			
Brand			
Keyboard type			
Brand x keybd			
Location		1500	
Division		1700	
Error			
Total		3375	

carry out Anova